

EXPERIMENT 30 : DESCRIPTION NUMERICAL PREDICTION ANALYSIS USING LINEAR REGRESSION THROUGH WEKA

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **Logistic -R 1.0E-8 -M -1 -num-decimal-places 4**

Test options:
☒ Use training set
☐ Supplied test set
☐ Cross-validation Folds: 10
☐ Percentage split %: 66
More options...

(Nom) class: Start Stop

Result list (right-click for options):
20:12:44 - functions.Logistic

Classifier output:

```

<math>\mu</math>=0.9994
<math>\lnu</math>=1.0012
<math>mas</math>=0.9142
<math>ped</math>=0.3886
<math>age</math>=0.9852

```

Time taken to build model: 0.02 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances	601	79.2552 %
Incorrectly Classified Instances	167	21.7448 %
Kappa statistic	0.4966	
Mean absolute error	0.3063	
Root mean squared error	0.3508	
Relative absolute error	67.3928 %	
Root relative squared error	81.9907 %	
Total Number of Instances	768	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
Weighted Avg.	0.890	0.418	0.799	0.890	0.842	0.504	0.839	0.897	tested_negative
	0.582	0.110	0.739	0.582	0.651	0.504	0.839	0.730	tested_positive

=== Confusion Matrix ===

	a	b	<-- classified as
445 55	a = tested_negative		
112 156	b = tested_positive		

Status: OK

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **Logistic -R 1.0E-8 -M -1 -num-decimal-places 4**

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds: 10
☐ Percentage split %: 66
More options...

(Nom) class: Start Stop

Result list (right-click for options):
20:12:44 - functions.Logistic
20:13:20 - functions.Logistic

Classifier output:

```

<math>\mu</math>=0.9994
<math>plas</math>=0.9654
<math>pres</math>=1.0134
<math>skin</math>=0.9994
<math>\lnu</math>=1.0012
<math>mas</math>=0.9142
<math>ped</math>=0.3886
<math>age</math>=0.9852

```

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	593	77.2135 %
Incorrectly Classified Instances	175	22.7865 %
Kappa statistic	0.4734	
Mean absolute error	0.3094	
Root mean squared error	0.3954	
Relative absolute error	68.0818 %	
Root relative squared error	82.9651 %	
Total Number of Instances	768	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
Weighted Avg.	0.880	0.429	0.793	0.880	0.834	0.480	0.832	0.892	tested_negative
	0.571	0.120	0.718	0.571	0.636	0.480	0.832	0.715	tested_positive

=== Confusion Matrix ===

	a	b	<-- classified as
440 60	a = tested_negative		
115 153	b = tested_positive		

Status: OK

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier: Choose **Logistic -R 1.0E-8 -M -1 -num-decimal-places 4**

Test options:
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds: 15
☐ Percentage split %: 66
More options...

(Nom) class: Start Stop

Result list (right-click for options):
20:12:44 - functions.Logistic
20:13:20 - functions.Logistic
20:14:02 - functions.Logistic

Classifier output:

```

<math>\mu</math>=0.9994
<math>plas</math>=0.9654
<math>pres</math>=1.0134
<math>skin</math>=0.9994
<math>\lnu</math>=1.0012
<math>mas</math>=0.9142
<math>ped</math>=0.3886
<math>age</math>=0.9852

```

Time taken to build model: 0.02 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	596	77.6042 %
Incorrectly Classified Instances	172	22.3958 %
Kappa statistic	0.4801	
Mean absolute error	0.3094	
Root mean squared error	0.3955	
Relative absolute error	68.0753 %	
Root relative squared error	82.9687 %	
Total Number of Instances	768	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
Weighted Avg.	0.888	0.433	0.793	0.888	0.838	0.488	0.832	0.891	tested_negative
	0.567	0.112	0.731	0.567	0.639	0.488	0.832	0.713	tested_positive

=== Confusion Matrix ===

	a	b	<-- classified as
444 56	a = tested_negative		
116 152	b = tested_positive		

Status: OK

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier Choose **Logistic -R 1.0E-8 -M -1 -num-decimal-places 4**

Test options
☐ Use training set
☐ Supplied test set Set...
☒ Cross-validation Folds 20
☐ Percentage split % 66
 More options...

(Nom) class

Result list (right-click for options)
 20:12:44 - functions.Logistic
 20:13:20 - functions.Logistic
 20:14:02 - functions.Logistic
20:14:40 - functions.Logistic

Classifier output

```

class
plas 0.9841
prea 0.9654
skin 1.0134
insu 0.9994
mass 1.0012
pedi 0.9142
age 0.3886
    0.9852

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances      595      77.474 %
Incorrectly Classified Instances    173      22.526 %
Kappa statistic                    0.4776
Mean absolute error                 0.3096
Root mean squared error             0.3557
Relative absolute error             68.1261 %
Root relative squared error         83.0152 %
Total Number of Instances          768

=== Detailed Accuracy By Class ===
               TP Rate  PP Rate  Precision  Recall  F-Measure  MDC   ROC Area  PRC Area  Class
0.886   0.433   0.752   0.886   0.837   0.485   0.832   0.891   tested_negative
0.567   0.114   0.727   0.567   0.637   0.485   0.832   0.713   tested_positive
Weighted Avg.   0.775   0.322   0.770   0.775   0.767   0.485   0.832   0.829

=== Confusion Matrix ===
      a  b  <-- classified as
443  57 |  a = tested_negative
116 152 |  b = tested_positive
  
```

Status
OK

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier Choose **LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4**

Test options
☒ Use training set
☐ Supplied test set Set...
☐ Cross-validation Folds 20
☐ Percentage split % 66
 More options...

(Num) age

Result list (right-click for options)
 20:12:44 - functions.Logistic
 20:13:20 - functions.Logistic
 20:14:02 - functions.Logistic
 20:14:40 - functions.Logistic
20:15:44 - functions.LinearRegression

Classifier output

```

class
insu
mass
pedi
age
class

Test mode: evaluate on training data

=== Classifier model (full training set) ===

Linear Regression Model

age =
1.6654 * preg +
0.0573 * plas +
0.1034 * prea +
-0.0911 * skin +
1.4283 * class=tested_positive +
14.1346

Time taken to build model: 0.1 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0.01 seconds

=== Summary ===
Correlation coefficient      0.6078
Mean absolute error          4.7393
Root mean squared error      9.3323
Relative absolute error      70.3002 %
Root relative squared error   79.4063 %
Total Number of Instances    768
  
```

Status
OK

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier Choose **LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4**

Test options
☒ Use training set
☐ Supplied test set Set...
☐ Cross-validation Folds 20
☐ Percentage split % 66
 More options...

(Num) skin

Result list (right-click for options)
 20:12:44 - functions.Logistic
 20:13:20 - functions.Logistic
 20:14:02 - functions.Logistic
 20:14:40 - functions.Logistic
 20:15:44 - functions.LinearRegression
20:16:30 - functions.LinearRegression

Classifier output

```

class
mass
pedi
age
class

Test mode: evaluate on training data

=== Classifier model (full training set) ===

Linear Regression Model

skin =
-0.0717 * plas +
0.1079 * prea +
0.0544 * insu +
0.6097 * mass +
4.1753 * pedi +
-0.1421 * age +
0.6541

Time taken to build model: 0.01 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===
Correlation coefficient      0.5802
Mean absolute error          10.6762
Root mean squared error      12.9044
Relative absolute error      78.1586 %
Root relative squared error   81.4464 %
Total Number of Instances    768
  
```

Status
OK